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MILITARY SPECIFICATION

CAPACITORS, BY-PASS, RADIO-INTERFERENCE REDUCTION, PAPER DIELECTRIC, AC AND DC, (HERMETICALLY SEALED IN METALLIC CASES), GENERAL SPECIFICATION FOR

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the performance and general material requirements for by-pass, radio-interference-reduction, alternating-current (ac) and direct-current (dc), paper-dielectric capacitors, hermetically sealed in metallic cases, for use primarily in broadband, radio-interference suppression application. In addition, this specification indicates the ambient test conditions within which the capacitors must operate satisfactorily and reliably. These capacitors are suitable for operation over a temperature range of —55° to +85°C.

1.2 Classification.

1.2.1 *Type designation.* The type designation of capacitors shall be in the following form, and as specified (see 3.1 and 6.2):

CA32	K	F	U	103
Style (1.2.1.1)	Terminal (1.2.1.2)	Characteristic (1.2.1.3)	Voltage (1.2.1.4)	Capacitance (1.2.1.5)

1.2.1.1 *Style.* The style is identified by the two-letter symbol "CA" followed by a two-digit number; the letters identify by-pass, radio-interference-reduction, ac and dc, paper-dielectric capacitors, hermetically sealed in metallic cases; the first digit identifies the general shape of the case and the second digit identifies specific details other than case size. Each style designation may include a family of case sizes.

FSC 5910

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1.2.1.2 *Terminals.* The terminal is identified by a single letter in accordance with table I.

TABLE I. *Terminal*

Symbol	Type of terminal
A	Axial-wire lead
B	Solder lug
K	Screw-type terminal

1.2.1.3 *Characteristic.* The characteristic is identified by a single letter in accordance with table II.

TABLE II

Characteristic	Values for characteristic F
High ambient test temperature, degrees centigrade ±3° C.	85
Low ambient test temperature, degrees centigrade ±3° C.	—55
"Megohms times microfarads":	
At 25° C.	1,500
At high ambient test temperature	15
Insulation resistance, in megohms:	
At 25° C.[1]	4,500
At high ambient test temperature	450
Capacitance change at —55° ± 3° C. from value at room ambient temperature, in percent	30
AC–DC capacitors:	
Life-test voltage, in percent of rated ac voltage rating of 250 volts or less, and volt-ampere rating of 200 or less	140
DC capacitors:	
Life-test voltage, in percent of rated dc voltage rating	140

[1] For corrections, if measurements are made at other temperatures between 20° and 40° C., see table III

1.2.1.4 *Voltage.* The voltage rating is identified by a single letter in accordance with table IV.

1.2.1.5 *Capacitance.* The nominal capacitance value expressed in micro-microfarads is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

TABLE III. *Insulation-Resistance Correction Factors*

Degrees centigrade	Correction factor[1]	Degrees centigrade	Correction factor[1]
20	1.42	31	0.67
21	1.33	32	.63
22	1.24	33	.59
23	1.16	34	.55
24	1.08	35	.51
25	1.00	36	.47
26	0.94	37	.44
27	.87	38	.41
28	.82	39	.38
29	.76	40	.36
30	.71		

[1] These correction factors are to be applied to the required values, not to the measured values.

TABLE IV. *Voltage rating*

Symbol	Volts, ac (rms) (up to and including 60 cps)	Volts, dc
B	...	100
U	250	250
W	500	500

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

VV-L-791 — Lubricants, Liquid Fuels, and Related Products; Method of Inspection, Sampling, and Testing.

PPP-B-566 — Boxes, Folding, Paperboard.

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PPP-B-575 — Box, Paper-Overlaid Veneer (Strap-around Type).

PPP-B-585 — Boxes; Wood, Wire-bound.

PPP-B-591 — Boxes; Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated-Plywood.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes, Fiber.

PPP-B-676 — Boxes, Set-Up, Paperboard.

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MIL-P-116 — Preservation, Methods of.

MIL-B-4229 — Boxes; Paperboard, Metal-Stayed.

MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case Waterproof.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-130 — Identification Marking of U.S. Military Property.

MIL-STD-202 — Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 — Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Detail requirements for individual capacitor styles. Detail requirements or exceptions applicable to particular styles of capacitors shall be as specified in the applicable detail specifications. In the event of any conflict between requirements of this specification and the detail specifications, the latter shall govern. (See 6.1.)

3.2 Qualification. Capacitors furnished under this specification shall be a product which has passed the qualification inspection specified in 4.4. (See 6.3 and 6.3.1.)

3.3 Material. The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 *Impregnating and filling compounds.*

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Compounds used in the impregnation and filling of capacitors shall be chemically inactive with respect to the capacitor element and the case. (See 3.4.1, 6.5, and 6.6). The compound, either in the state of original application or as a result of having aged, shall have no adverse effect on the performance of the capacitor. For liquid-filled capacitors, the same material shall be used for impregnating as is used for filling.

3.4 **Design and construction.** Capacitors shall be of the design, construction, and physical dimensions specified. (See 3.1.)

3.4.1 *Case.* Each capacitor shall be enclosed in an hermetically sealed metallic case which will prevent leakage of the impregnant or filling, and in addition, will protect the capacitor element from moisture and mechanical damage under all test conditions specified herein.

3.4.2 *Finish.* All exposed metallic surfaces shall be suitably protected against corrosion by plating, or by other means. (See 3.12.) The resultant finish shall form a good electrical conductor and shall be free from defects that may affect its protective value.

3.4.3 *Threaded parts.* All threaded parts shall be in accordance with Handbook H28. Wherever practical, all threads shall be in conformity with the coarse-thread series. The fine-thread series shall be used only for applications that might show a definite advantage through their use. Where a special diameter-pitch combination is required, the thread shall be of American National Form and of any pitch between 16 and 36 which is used in the fine-thread series. Aluminum nuts shall not be used.

3.5 **Terminal strength.** As a result of the test specified in 4.6.2, no part of the terminal shall loosen or rupture, and there shall be no other damage. Screw-type terminals shall exhibit no perceptible movement relative to the case, under the applied torque.

3.6 **Dielectric strength.** Capacitors shall withstand the potential specified in 4.6.3 without visible damage, flashover, breakdown, or open-or short-circuiting.

3.7 **Insulation resistance.** When the product of the nominal capacitance expressed in microfarads (μf) and the insulation resistance expressed in megohms is equal to or greater than the applicable value shown in table II at the specified temperature, the insulation resistance need not exceed the value shown in table II. However, when this product is less than the applicable value specified in table II, the insulation resistance shall meet the value shown on figure 1, at the specified temperature. The insulation resistance shall at no time fall below the required value, regardless of the test voltage used. (See 4.6.4.)

3.8 **Capacitance.** When measured as specified in 4.6.5, the capacitance shall be within the specified tolerance. (See 3.1.)

3.9 **Dissipation factor.** When measured as specified in 4.6.6, the dissipation factor shall be not more than 1.0 percent.

3.10 **Seal.** When capacitors are tested as specified in 4.6.7, there shall be no evidence of leakage.

3.11 **Vibration.** As a result of the test specified in 4.6.8, there shall be no mechanical damage to the capacitor, and no evidence of intermittent contacts, or open-or short-circuiting.

3.12 **Corrosion.** As a result of the test specified in 4.6.9, there shall be no harmful or extensive corrosion and at least 90 percent of any exposed metallic surfaces of the capacitor shall be protected by the finish. (See 3.4.2.) In addition, there shall be not more than 10 percent corrosion of the terminal hardware or mounting surface. Marking shall remain legible after the test.

3.13 **Temperature and immersion cycling.**

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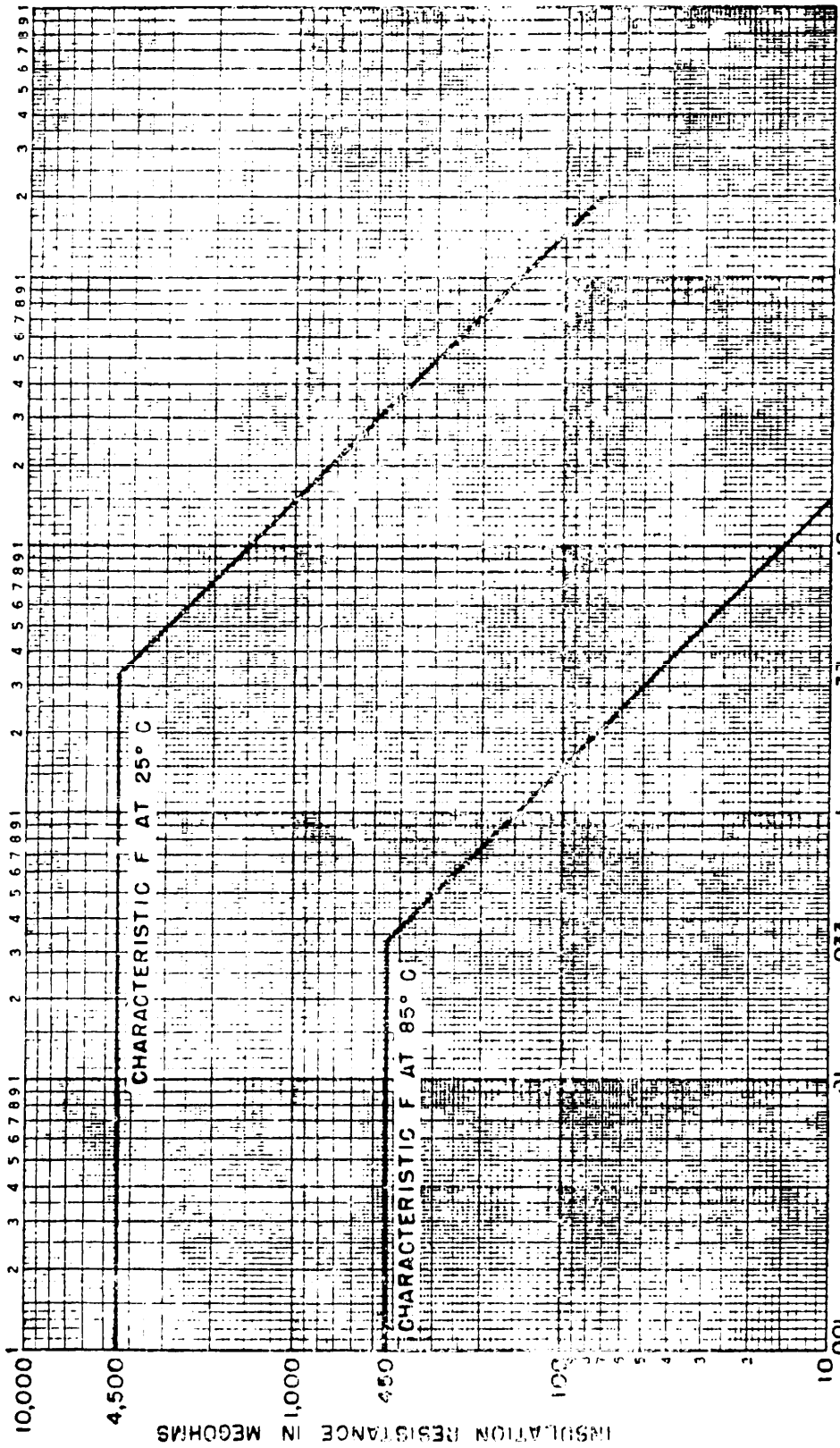


FIGURE 1. *Graphical representation of insulation-resistance requirements specified.* (See table II and 3.7.)

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When tested as specified in 4.6.10, capacitors shall meet the following requirements:

Dielectric strength	As specified in 3.6.
Insulation resistance (at 25° C.)	Not less than 75 percent of the value specified in 3.7.
Capacitance	Change not more than 5 percent of the nominal value from the initial value obtained when measured as specified in 4.6.5.
Dissipation factor	Shall not exceed 1.5 percent.
Corrosion	As specified in 3.12.

3.14 Moisture resistance. When tested as specified in 4.6.11, capacitors shall meet the following requirements:

Dielectric strength	As specified in 3.6.
Insulation resistance (at 25° C.)	Not less than 75 percent of the value specified in 3.7.
Capacitance	Change not more than 5 percent of the nominal value from the initial value obtained when measured as specified in 4.6.5.
Dissipation factor	Shall not exceed 1.5 percent.
Corrosion	As specified in 3.12.

3.15 Low ambient temperature and capacitance change at reduced temperature. When tested as specified in 4.6.12, capacitors shall withstand the application of rated voltage (see 3.1) without breakdown or flashover. The capacitance change at —55° ±3°C. shall not exceed the limit shown in table II.

3.16 Life. When tested as specified in 4.6.13, capacitors shall meet the following requirements:

Insulation resistance (at 25° C.)	Not less than 75 percent of the value specified in 3.7.
Capacitance	Change not more than 10 percent of the nominal value from the initial value obtained when measured as specified in 4.6.5.
Dissipation factor	

There shall be no mechanical failure, leakage of impregnant or filling compound, nor open- or short-circuiting.

3.17 Flashpoint of impregnant or filling compound. When measured as specified in 4.6.14, the flashpoint of the impregnant or filling compound shall be not lower than 135°C. This shall apply to liquid impregnants and filling compounds and to solid impregnants having a melting point of less than 135°C.

3.18 Marking. Capacitors shall be marked in accordance with Standard MIL-STD-130, with the type designation, the manufacturer's name or symbol and the voltage rating. There shall be no space between the symbols which comprise the type designation. If space is limited, the voltage rating may be omitted and the type designation may appear on two lines. In this event, the type designation shall be divided between the terminal symbol and the characteristic symbol as shown in the following example:

CA32K
FU103

Marking shall remain legible after all tests.

3.19 Workmanship. Capacitors shall be manufactured and processed in a careful and workmanlike manner, in accordance with good design and sound practice, and shall meet all requirements of this specification.

3.19.1 *Soldering.* Flux for soldering shall be rosin, rosin and alcohol, or rosin and turpentine. No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material, unless they are removed in the processing. Where acid or acid salts are used, as permitted above, they shall be completely

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neutralized and removed immediately after use. All excess flux and solder shall be removed. Electrical connections shall be electrically continuous after soldering and where practicable, shall be mechanically secured before soldering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of inspection. The examination and testing of capacitors shall be classified as follows:

(a) Qualification inspection. (See 4.4.)

(b) Acceptance inspection. (See 4.5.)

4.2 Inspection conditions. Unless otherwise specified herein, all inspection shall be made at room ambient temperature, pressure, and humidity.

4.3 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required acceptance inspection. The manufacturer shall establish adequate calibration of test equipment to the satisfaction of the Government.

4.4 Qualification inspection.

4.4.1 *Sample.* The number of specimens comprising a sample of capacitors and the amount of impregnant or filling compound to be submitted for qualification inspection shall be as specified in the appendix to this specification.

4.4.2 *Inspection routine.* The specimens will be subjected to the qualification inspection specified in table V, in the order shown. Two specimens in a sample will be subjected to visual and mechanical examination (internal). The remaining specimens will be subjected to the subsequent inspection of group I. The specimens will then be divided into the three remaining groups shown in table V, and subjected to the inspection for their particular group.

4.4.3 *Defectives.* Defectives in excess of those allowed in table V will be cause for refusal to grant qualification.

4.5 Acceptance inspection. Acceptance inspection shall consist of groups A, B, and C.

4.5.1 *Inspection lot.* An inspection lot shall be as specified in Standard MIL-STD-105, and shall consist of capacitors of the same style, characteristic, and voltage, presented at the same time.

4.5.2 *Resubmitted lots.* If an inspection lot is rejected, the manufacturer may rework the lot or screen out defectives and resubmit it for acceptance inspection. Resubmitted lots shall be kept separate from new lots. The resubmitted lots shall be inspected, using tightened inspection.

4.5.3 *Group A inspection.* Group A inspection shall consist of the examinations and tests specified in table VI. Statistical sampling and inspection shall be in accordance with Standard MIL-STD-105. The acceptable quality levels (AQL) shall be as specified in table VI.

4.5.4 *Group B inspection.* Group B inspection shall consist of the tests specified in table VII, in the order shown.

4.5.4.1 *Sampling procedure.* The sampling procedure shall be in accordance with the appendix to Standard MIL-STD-105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The reduced inspection procedure shall be R-1. The AQL shall be 4.0 (percent defective) and the inspection level shall be L8 for normal inspection and L6 for reduced inspection.

4.5.4.2 *Disposition of sample units.* Sample units which have passed all group B inspection shall be delivered on the contract or order, if the lot is accepted.

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TABLE V. *Qualification Inspection*

Examination or test	Requirement paragraph	Method paragraph	Number of specimens to be inspected	Number of defectives allowed[1]
Flashpoint of impregnant or filling compound	3.17	4.6.14	...	0
Group I				
Visual and mechanical examination (internal):				
Material, design, construction, and workmanship	3.1, 3.3 to 3.4.3, incl, 3.19 and 3.19.1	4.6.1	2	
Visual and mechanical examination (external):				
Physical dimensions, marking,[2] and workmanship	3.4, 3.18 to 3.19.1, incl	4.6.1		1
Terminal strength	3.5	4.6.2		
Dielectric strength	3.6	4.6.3.1		
Insulation resistance	3.7	4.6.4.1	[3] 25	
Capacitance	3.8	4.6.5		
Dissipation factor	3.9	4.6.6		
Seal	3.10	4.6.7		
Group II				
Vibration	3.11	4.6.8		
Corrosion	3.12	4.6.9	6	
Temperature and immersion cycling	3.13	4.6.10		
Group III				1
Moisture resistance	3.14	4.6.11	6	
Group IV				
Low ambient temperature and capacitance change at reduced temperature	3.15	4.6.12	12	1
Life	3.16	4.6.13		

[1] A specimen having one or more defects will be considered as a single defective.

[2] Marking will be considered as a defect only if it becomes illegible as a result of any of the tests of the specification.

[3] One additional specimen is included in each sample of 27 specimens to permit substitution for the allowable defective in group I.

TABLE VI. *Group A Inspection*

Examination or test	Requirement paragraph	Method paragraph	Defect classification[1]	AQL (percent defective)	
				Major	Minor
Visual and mechanical examination:					
Material	3.1, 3.3 and 3.3.1	4.6.1	Major		
Body dimensions	3.4	...	Major		
Design and construction (other than body dimensions)	3.4 to 3.4.3, incl	...	Minor	1.0	4.0
Marking[2]	3.18	...	Major		
Workmanship	3.19 and 3.19.1	...	Minor		
Dielectric strength	3.6	4.6.3.2	Major		
Insulation resistance (at 25° C.)	3.7	4.6.4.2	Major		
Capacitance	3.8	4.6.5	Major	1.0	...
Dissipation factor	3.9	4.6.6			
Seal	3.10	4.6.7	Major		

[1] Major and minor defects shall be as defined in Standard MIL-STD-105.

[2] Marking defects are based on visual examination only and shall be charged only for illegible, incomplete, or incorrect markings. Any subsequent electrical defects shall not be used as a basis for determining marking defects.

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TABLE VII. *Group B Inspection*

Test	Requirement paragraph	Method paragraph
Terminal strength	3.5	4.6.2
Insulation resistance (at high ambient test temperature)	3.7	4.6.4.2
Vibration	3.11	4.6.8
Temperature and immersion cycling	3.13	4.6.10

4.5.5 *Group C inspection.* Group C inspection shall consist of the tests specified in table VIII, in the order shown for each subgroup. It shall be performed on sample units that have passed the group A inspection. The number of sample units to be tested for each subgroup shall be as specified in table VIII. Separate sample units shall be used for each subgroup.

4.5.5.1 *Sampling procedure.* Once each calendar month, sample units shall be selected according to sytle, irrespective of the characteristic, voltage, and capacitance value. Shipments of lots shall not be held up pending results of group C inspection, except as specified in 4.5.5.3.

TABLE VIII. *Group C Inspection*

Test	Requirement paragraph	Method paragraph	Number of sample units to be tested	Number of defectives allowed[1]
Subgroup 1				
Corrosion	3.12	4.6.9	6	1
Subgroup 2				
Moisture resistance	3.14	4.6.11	6	
Subgroup 3				
Low ambient temperature and capacitance change at reduced temperature	3.15	4.6.12	12	1
Life	3.16	4.6.13		

[1] A specimen having one or more defects will be considered a single defective.

4.5.5.2 *Disposition of sample units.* Sample units subjected to group C inspection shall not be delivered on the contract or order.

4.5.5.3 *Noncompliance.* If a sample fails to pass Group C inspection, the manufacturer shall take corrective action on the process and on all units of product which can be corrected and which were manufactured with the same conditions, materials, processes, etc., and are considered subject to the same test failure. Acceptance inspection shall be discontinued until corrective action has been taken. After the corrective action, sample units shall be subjected to the necessary group C inspection (all tests, or the failed tests at the option of the Government). Groups A and B inspection may be reinstituted; however, final acceptance shall be withheld until the group C inspection has shown that the corrective action was successful.

4.6 Methods of examination and test.

4.6.1 *Visual and mechanical examination.* Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements. (See 3.1, 3.3 to 3.4.3, incl., and 3.18 to 3.19.1, incl.)

4.6.2 *Terminal strength* (see 3.5).

4.6.2.1 *Pull.* A force of 5 pounds shall be applied to the terminal at the point where it is normally connected in an actual circuit,

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except that for axial-wire leads, the contact shall be made 5/8 inch from the insulator (case). The force shall be applied axially and shall be increased gradually to the specified magnitude and held at that value for at least 30 seconds.

4.6.2.2 *Torque (tapped, screw-type terminals only).* The capacitor case shall be held rigid while the applicable torque specified in table IX is applied radially to the screw. The torque shall be applied in the direction which will tighten the screw and shall then be held at its specified value for a period of not less than 3 seconds for each terminal. It shall be applied alternately to the terminals until a total of five twists have been applied to each terminal.

TABLE IX. *Torque*

Screw size	Torque (pound-inches)
8-32	14
10-32	18
1/4-20	30
5/16-24	50

4.6.3 *Dielectric strength* (see 3.6).

4.6.3.1 *For qualification inspection.* A potential of 200 percent of the rated dc voltage for dc rated capacitors and 300 percent of the rated dc voltage for ac-dc rated capacitors, shall be applied to the capacitor for a period of 2 minutes ± 5 seconds. The potential shall be applied between the terminals and the case. For dc capacitors, the ac ripple shall not exceed 2 percent. The surge current shall be limited to between 5 milliamperes and 1 ampere. When necessary, a suitable current-limiting resistor shall be inserted into the circuit. Not more than 5 percent of the generator voltage shall appear across the limiting resistor of the capacitor during the application of the specified potential. A voltmeter shall be used to measure the applied voltage to an accuracy of at least 5 percent.

4.6.3.2 *For acceptance inspection.* Capacitors shall be tested as specified in 4.6.3.1 except that the accuracy of the voltmeter need be not greater than ± 5 percent. However, at the discretion of the manufacturer, a test voltage of 250 percent of the rated dc voltage may be applied for a period of not less than 1 second.

4.6.4 *Insulation resistance* (see 3.7).

4.6.4.1 *For qualification inspection.* Insulation resistance at the high ambient test temperature (see table II) and 25°C, or corrected thereto, shall be measured between the terminal and case, at a dc potential of 100 ± 10 volts for capacitors rated at 400 volts or under and a dc potential of 500 ± 50 volts for capacitors rated at over 400 volts. The electrification time shall be 2 minutes ± 5 seconds, and the accuracy of equipment shall be within ± 10 percent for measurements within the required range.

4.6.4.2 *For acceptance inspection.* Capacitors shall be tested as specified in 4.6.4.1, except that the test potential may be 100 volts for all voltage ratings. In addition, at the option of the manufacturer, the electrification time may be the time required by the capacitor to reach full charge, provided it does not exceed 2 minutes.

4.6.5 *Capacitance.* Capacitance shall be measured at a frequency of 1,000 ± 100 cycles per second (cps). Measurement accuracy shall be within ±2 percent. (See 3.8.)

4.6.6 *Dissipation factor.* The dissipation factor of each capacitor shall be measured at an a-c, root-mean-square voltage not greater than 20 percent of the rated voltage. (See table IV.) Measurement shall be made at a frequency of 1,000 ± 100 cps. Measurement accuracy shall be within ± 2 percent. (See 3.9.)

4.6.7 *Seal* (see 3.10).

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4.6.7.1 *Solid-filled and liquid-impregnated capacitors.* Capacitors whose filling material will be solid at the test temperature, shall be immersed for not less than 1 minute in oil or water maintained at a temperature of 95° $^{+5°}_{-0°}$ C. If no filling material is used, this test shall also apply to liquid-impregnated capacitors. (See 6.5.)

4.6.7.2 *Liquid-filled capacitors.* Capacitors whose filling material will be liquid at the test temperature, shall be placed with the terminals facing sideways and brought to a case temperature of 95° $^{+5°}_{-0°}$ C. After 10 minutes at this case temperature the capacitors shall be turned through 180° onto another surface with the terminals still sideways and kept in this position for 10 minutes with the case temperature maintained within the same limits. (See 6.6.)

4.6.8 *Vibration (see 3.11).* Capacitors shall be tested in accordance with method 201 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting — Securely fastened by normal mounting means (see 3.1).

(b) Test condition — B, except that for acceptance inspection the duration shall be 1½ hours.

(c) Initial measurements — Not applicable.

(d) Motion — For the first 1¼ hours, motion shall be applied in the direction vertical to the cylindrical axis; for the remaining time, motion shall be applied in the direction parallel to the cylindrical axis, except that for acceptance inspection, motion may be applied in either of the 2 specified directions.

(e) Final measurements — During the last ½ hour of vibration, an electrical measurement shall be made to determine intermittent contacts, or open-or short-circuiting.

4.6.9 *Corrosion.* Capacitors shall be tested in accordance with method 101, test condition B, of Standard MIL-STD-202. After this test, capacitors shall be visually examined for evidence of corrosion. (See 3.12.)

4.6.10 *Temperature and immersion cycling* (see 3.13).

4.6.10.1 *Temperature cycling.* Capacitors shall be tested in accordance with method 102, test condition A, of Standard MIL-STD-202, except that no measurements shall be made before and after cycling.

4.6.10.2 *Immersion cycling.* Within 4 to 24 hours after completion of temperature cycling, capacitors shall be tested in accordance with method 104 of Standard MIL-STD-202. The following details and exception shall apply:

(a) Test condition — B.

(b) Measurements — Within 2 hours after capacitors are removed from the immersion tank, dielectric strength, insulation resistance at 25°C., capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, 4.6.5, and 4.6.6, respectively. If the capacitors fail the insulation-resistance requirement, they may be retested for insulation resistance after a total maximum drying period of 24 hours.

(c) Visual examination — Capacitors shall be examined for corrosion and obliteration of marking.

4.6.11 *Moisture resistance* (see 3.14). Capacitors shall be tested in accordance with method 102, test condition A, of Standard MIL-STD-202, except that no measurements shall be made before and after cycling. Capacitors shall then be tested in accordance with method 106 of Standard MIL-STD-202. The following details and exceptions shall apply:

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(a) Mounting — Securely fastened by normal mounting means (see 3.1) except during measurements.

(b) Conditioning prior to first cycle — Not applicable.

(c) Initial measurements — Not applicable.

(d) Polarization — During steps 1 to 6 inclusive, a dc potential of 100 volts shall be applied to 50 percent of the capacitors. No potential shall be applied to the remaining 50 percent of the capacitors.

(e) Final measurements — Upon completion of step 6 of the final cycle, the dielectric strength, insulation resistance at 25°C., capacitance, and dissipation factor shall be measured as specified in 4.6.3, 4.6.4, 4.6.5, and 4.6.6, respectively, at a relative humidity of 50 ± 5 percent.

(f) Visual examination — Capacitors shall be examined for corrosion and obliteration of marking.

4.6.12 *Low ambient temperature and capacitance change at reduced temperature* (see 3.15).

4.6.12.1 *Low ambient temperature.* Capacitors shall be placed in a chamber maintained at —55° ± 3°C., and rated dc voltage shall be applied at this condition for 48 ± 4 hours. (See 3.1.) The air within the conditioning chamber shall be circulated.

4.6.12.2 *Capacitance change at reduced temperature.* At the conclusion of the test specified in 4.6.12.1, but before the capacitors are removed from the conditioning chamber, capacitance shall be measured as specified in 4.6.5, except that the measurement shall be made at —55° ± 3°C. This measurement shall be compared with a measurement made at room ambient temperature.

4.6.13 *Life.* Capacitors shall be subjected to the life-test voltage specified in table II, ac or dc, whichever is applicable, at the high ambient test temperature specified in table II for 250 ± 8 hours. The potential shall be applied between the terminal and the case for the duration of the test. During the test, the capacitors shall be suspended by their terminals within a test chamber and separated from each other by a distance of not less than 1½ inches, to allow free circulation of air around all surfaces of each capacitor. While in the test chamber, capacitors shall be mantained at high ambient test temperature (see table II) throughout the test. At the conclusion of the test, capacitors shall be returned to the test condiions specified in 4.2 and shall be visually examined for leakage of impregnant or filling compound, and deformation of case. Insulation resistance at 25°C., capacitance, and dissipation factor shall then be measured as specified in 4.6.4, 4.6.5, and 4.6.6, respectively. (See 3.16.)

4.6.14 *Flashpoint of impregnant or filling compound.* The flashpoint of impregnant or filling compound shall be measured as specified in method 1103.5 of Specification VV-L-791, except that fire point and precision do not apply. (See 3.17.)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.2).

5.1.1 *Level A.* Capacitors shall be individually protected and unit-packaged in accordance with method III of Specification MIL-P-116. Unless otherwise specified (see 6.2), five unit packages or a multiple thereof shall be further packaged in intermediate containers conforming to Specification PPP-B-566, PPP-B-676, or MIL-B-4229. The gross weight of the intermediate container shall not exceed 10 pounds.

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5.1.2 *Level C.* Capacitors shall be afforded preservation and packaging in accordance with the manufacturer's normal commercial practice.

5.2 Packing. (see 6.2).

5.2.1 *Level A.* Capacitors packaged as specified (see 6.2) shall be packed in type II straparound, wirebound wood (for class 3 use), wood-cleated fiberboard, wood-cleated plywood (overseas type), nailed wood, corrugated or solid fiberboard (class 2, grade 3 or better), or wood-cleated paper overlaid (overseas type) boxes conforming to Specifications PPP–B–575, PPP–B–585, PPP–B–591, PPP–B–601, PPP–B–621, PPP–B–636, and MIL–B–10377, respectively. Shipping containers shall have case liners conforming to Specification MIL–L–10547 and appendix thereto. Case liners for boxes conforming to Specification PPP–B–636 may be omitted provided all joints of the boxes are sealed with tape as specified in the box specification. Box closures shall be as specified in the appendix of the applicable box specification. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the box specification.

5.2.2. *Level B.* Capacitors packaged as specified (see 6.2) shall be packed in domestic-type wirebound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, corrugated or solid fiberboard, or wood-cleated paper-overlaid boxes conforming to Specifications PPP–B–585, PPP–B–591, PPP–B–601, PPP–B–621, PPP–B–636, and MIL–B–10377, respectively. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the box specification.

5.2.3 *Level C.* Capacitors packaged as specified (see 6.2) shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Consolidated Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.2.4 *General.* Insofar as possible and practical, exterior containers shall be uniform in shape and size, and shall be of minimum cube and tare consistent with the protection required.

5.3 **Marking.** In addition to any special marking required by the contract or order, unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with Standard MIL–STD–129. (See 6.2.)

6. NOTES

6.1 **Intended use.** The capacitors covered by this specification are intended for general purpose application where suppression of broadband radio interference is desired.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of the applicable detail specification, and the complete type designation. (See 1.2.1 and 3.1.)

(c) Levels of preservation and packaging and packing, and applicable marking. (See sect 5.)

(d) Number of unit packages if other than that specified in 5.1.1.

6.2.1 *Indirect shipments.* The packaging, packing, and marking specified in section 5 apply only to direct purchases by or direct shipments to the Government and are not intended to apply to contracts or orders be-

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tween the manufacturer and prime contractor.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the ...d opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Armed Services Electro-Standards Agency (ASESA), Fort Monmouth, N. J.

6.4 Radio interference. Radio interference is defined as undesired, conducted or radiated electrical disturbances, including transients, within the range of frequencies covered by this specification, which may interfere with the operation of electrical or electronic communications equipment or other electronic equipment.

6.5 Liquid-impregnated capacitor. A liquid-impregnated capacitor is a capacitor in which a liquid impregnant is dominantly contained within the foil-and-paper winding, but does not occupy substantially all of the case volume not required by the capacitor element and its connections.

6.6 Liquid-filled capacitor. A liquid-filled capacitor is a capacitor in which a liquid impregnant occupies substantially all of the case volume not required by the capacitor element and its connections. Space may be allowed for the expansion of the liquid under temperature variations.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodians:
Army—Signal Corps
Navy—Bureau of Ships
Air Force

Other interests:
Army—EO
Navy—AMCOr

Preparing activity:
Army—Signal Corps

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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 This appendix details the procedure for submission of samples, with related data, for qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined herein.

20. SUBMISSION

20.1 **Sample.**

20.1.1 *Single-type submission.* A sample consisting of 27 specimens of the type for which qualification is sought shall be submitted. (See 30.1.)

20.1.2 *Combined-type submission.* A maximum of 2 types of capacitors, 14 of each type, may be represented in a combined submission. Styles are grouped as shown in table X. Both types shall have the same impregnant. (See 30.2.)

20.1.3 *Impregnant.* A minimum of 200 cubic centimeters of each impregnant or filling compound used in the capacitors on a certification of the flashpoint of impregnant or filling compound as determined in accordance with 4.6.14, shall be submitted.

20.2 **Test data.** When required by ASESA, each submission shall be accompanied by test data covering the nondestructive tests listed in table V which have been performed on the submitted specimens. The performance of the destructive tests by the manufacturer on a duplicate set of specimens is encouraged, although not required. All test data shall be submitted in duplicate.

20.3 **Certification of material.** When submitting samples for qualification, the manufacturer shall submit certification, in duplicate, that the materials used in his components are in accordance with the applicable specification requirements.

20.4 **Description of products.** The manufacturer shall submit a detailed description of the materials and constructional features of the capacitors being submitted for test, including information on whether they are liquid filled or liquid impregnated; the type and quantity of the impregnant; the type, thickness, and number of layers of capacitor tissue and foil; the material, thickness, and applied finish of the case; and details of the terminal assemblies.

30. EXTENT OF QUALIFICATION

30.1 **Single-type submission.** Qualification will be restricted to the voltage ratings and capacitance values equal to and less than those of the type submitted.

30.2 **Combined-type submission.** Qualification of the types submitted will be a basis for qualification of the styles represented with the same provisions as stated in 30.1. An example of combined-type submission is shown in table X.

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TABLE X. *Example of Combined-Type Submission*

Styles	Types to be submitted	Number of specimens	Voltage rating	Capacitance value
			Volts	*μμf*
CA32, CA33, and CA34	CA32KFB504 or CA33KFB504, and CA34KFB504	14	100	0.5
CA36, CA37, and CA38	CA36KFW254 or CA37KFW254, and CA38KFW254	14	500	.25

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

FORM 1426
1 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

C-6279

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transferring 10 ml aliquots from each flask. Add 40 ml of glacial

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